Appendix A

Table A.1 Agro-climatic characteristics of study locations

	Umbarda Cluster 1	**Alasana Cluster 2**	**Garaj Cluster 3**	**Old Dawachwadi Cluster 4**	**Vijaynagar Cluster 5**	**Junoni Cluster 6**	**Yehala Cluster 7**
District	Washim	Buldhana	Aurangabad	Nasik	Satara	Osmanabad	Yavatamal
Block	Karanja	Shegaon	Vaijapur	Niphad	Satara	Osmanabad	Pusad
Villages	Umbarda Bz., Yevta, Manbha, Pimpri Warghat, Sukali, Dughora, Pilkheda, Somthana, Wadgaon	Alasana, Kherda, Ghoulkhed, Takali (viro), Takali (hat)	Garaj	Kumbhari, Panchkeshwar	Jihe, Phad-tarwadi, 25% of Angapur Vandan	Junoni, Walgud	Yehala, Kondhai
Geographical area (ha)	8505	2282	804	961	1214	1408	1280
Water sources	Open wells, borewell and irrigation tank	Open well	Open well, borewell	Open well, Storage tank, Canal	River, Open well	Open well, borewell, Irrigation tank	Dam, Open well
Surface water irrigation coverage	5%	0%	0%				
Crops	Soybean, Cotton, Pigeon Pea, Wheat, Gram, Orange and Lime	Soybean, Cotton, Gram, Wheat, Sugarcane, Pigeon Pea	Cotton, Pigeon Pea, Maize, Ginger, Gram, Wheat, Sweet orange	Grapes, Maize, Soybean, Ground-nut	Sugarcane Soybean, Rabi Sorghum	Soybean Sugarcane, Rabi Sorghum Wheat, Grapes	Soybean, Cotton, Pigeon Pea Wheat, Gram, Turmeric Groundnut
Rainfall Zone	Moderate rainfall zone	Assured rainfall zone	Scarcity rainfall zone	Scarcity rainfall zone	Scarcity rainfall zone	Assured rainfall zone	Moderate rainfall zone
Average. annual rainfall (mm)	684	682	541	564	650	953	780
Rainfall, year of analysis (mm)	671	746	993	410	551		

Table A.2 Division wise cropping data of Maharashtra 2019-20. Main crops are the 8 highest cultivated areas, except for rice which is unirrigated. The crops in the table cover 96% of the gross sown area of Maharashtra (*Areas in '000 ha*) *Source: Department of Agriculture, Government of Maharashtra*

	Main crops: Cover 79% of gross sown area of Maharashtra								**Unirrigated crops**							
	Cotton	Soybean	Gram	Rabi Sorghum	Pigeon Pea	Wheat	Sugar-cane	Kh. Maize	Rice	Mung bean	Black gram	Kh. Sorghum	Pearl Millet	Rb Maize	Major crops	Total crops
Konkan	0	0	4	0	3	0	0	0	366	0	4	0	0	0	6	376
Khandesh	956	154	228	72	68	240	35	430	110	71	62	79	172	88	2182	2763
Western Maharashtra	160	293	301	1307	84	261	593	144	254	116	80	82	350	137	3143	4161
Marathwada	1688	1928	818	578	503	207	174	136	3	134	139	170	149	56	6033	6683
West Vidarbha	1373	1540	549	22	528	207	13	60	6	63	55	55	3	15	4292	4488
East Vidarbha	315	210	143	3	133	142	8	2	739	2	1	4	0	3	955	1705
State total	4491	4124	2043	1982	1319	1057	822	772	1478	387	341	390	673	298	16610	20176

Section A.1 Representativeness of clusters

Table A.3 lists high level cropping and water sources data for the 5 clusters. Out of the 8 main crops from table A.2, a crop is termed 'major' for a division if greater than 20% of the crop in the state is cultivated in that division. In row 2 of table A.3 the denominator is the number of these major crops cultivated in the division. The numerator is the number of these major crops cultivated in the cluster. A large fraction indicates that the division cropping is well represented in the cluster.

Rows 3 and 4 indicate the horticultural cropping in the division and cluster respectively. Row 5 lists the type of water sources present in each cluster.

Table A.3 Field crops and horticultural crops representation of division in selected clusters, and type of water sources

Cluster	**1**	**2**	**3**	**4**	**5**
Division	**West Vidarbha**		**Marathwada**	**Khandesh**	**Western Maha.**
Major crops of the division in cluster / number of major crops in the division	5/5		4/7	2/3	3/3
Horticultural area in the division as % of horticultural area in state	9%		10%	33%	34%
Horticultural area in cluster as % of gross cropped area	8%	0	16%	57%	15%
Water sources	Open well, Borewell	Open well	Open well, Borewell	Open well, Storage tanks, Canal (69%)	Open well, River (34%)

Table A.4. Crops, irrigation methods, sources and transfers in the measurement clusters

Cluster	**1**	**3**	**6**	**7**
Number of major field crops	5	5	4	5
Number of horticultural crops	2	4	2	0
Water sources	Open well,	Open well,	Open well, Borewell,	Open well, Dam

	Borewell	Borewell	Irrigation tank (20%)	(50%)
Water transfers	Open well to borewell	Open well to borewell	Open well to borewell, Long distance pumping	Long distance pumping

Section A.2

The furrow or border lengths are 75 to 150 m. Border widths are 6 feet. Furrow spacings are 4 feet for most crops. The furrows for Pigeon pea are a little greater at 6 feet, and cultivated intercropped with soybean. Some crops such as cotton are sown in furrows, but after the monsoons the structure and irrigation becomes more border-like, which is when most of the irrigation takes place. These are included in the border method.

Table A.5 Number of irrigation events by crop and irrigation method for cluster 1, Umbarda, Washim

Crop	Average number of irrigations	Standard. Deviation	Minimum number of irrigations	Maximum number of irrigations	Season	Number of farmers surveyed
Wheat	9.1	2.4	6	14	Nov-Mar	42
Orange (Drip)	2 furrow (Nov-Dec) + 46 l/tree /day (avg. measured) Jan – May [a]				Multi year	26
Orange (Furrow)	2 furrow (Nov-Dec) + 135 l/ tree /day (estimated + measured) Jan – May [a]				Multi year	3
Mung bean, Black gram	Rainfed				June-Aug	
Soybean (10% Irrigated)	1.8	0.7	1	3	Jun-Sep	11
Cotton [b]	3	1.8	1	8	Jun- Nov	21
Cotton (drip)	8.5	5.8	3	20	Jun- Nov	7
Pigeon Pea	2.3	1.3	1	7	Jun-Jan	22
Gram (sprinkler)	4.1	1.8	2	8	Nov-Feb	8
Gram (border)	4	2	2	6	Nov-Feb	3

[a] The irrigation data is average measured data

[b] The cotton crop in 2020 -21 was harvested early between September and November instead of the usual January to February

Table A.6 Number of irrigations by crop and irrigation method for Cluster 3, Garaj, Aurangabad

Crop	Average number of irrigations	Standard. Deviation	Minimum number of irrigations	Maximum number of irrigations	Season	Number of farmers surveyed
Sweet Orange (Drip)	2 furrow (Nov-Dec) + 46 l/tree /day (avg. measured) Jan – May [a]				Multi year	26
Gram (border)	3.4	2.7	1	4	Dec-Mar	8
Rabi Sorghum	4.6	2.1	3	7	Nov-Mar	3
Ginger (drip)	17	2.3	15	20	Jun-May	7
Maize (kharif)	1	-	1	1	Jul-Oct	2
Pearl millet	5	1.4	3	6	Jul-Dec	4
Rabi Onion (border)	12	-	11	13	Jan-May	2
Rabi Onion (drip)	12	1.5	12	13	Jan-May	3
Cotton [b]	2	0.6	1	3	Jun-Sep	6
Cotton(drip) [b]	2	-	1	4	Jun-Sep	3

[a] The irrigation data is average measured data

[b]The cotton crop in 2020 -21 was harvested early between September and November instead of the usual January to February

Table A.7 Number of irrigations by crop and irrigation method for Cluster 2, Alasana, Buldhana

Crop	Average number of irrigations	Standard. Deviation	Minimum number of irrigations	Maximum number of irrigations	Season	Number of farmers surveyed
Cotton surface	6		1	13	Jun-Feb	4
Cotton drip	30		3	180	Jun-Feb	6
Soybean(10% irrigated)	1.5		1	4	Jun -Sep	11
Pigeon Pea	2		3	5	Jun - Jan	4
Gram	2				Nov-Mar	FGD [c]
Sugarcane	36				Jun-May	FGD
Wheat	6				Nov-Mar	FGD

[c] Focused Group Discussions

Table A.8 Number of irrigations by crop and irrigation method for Cluster 4, Old Dawachwadi , Nasik

Crop	Average number of irrigations	Standard. Deviation	Minimum number of irrigations	Maximum number of irrigations	Season	Number of farmers surveyed
Grape drip [d]			103	205	Multi year	4
Rabi Onion	10		9	10	Dec-Mar	4
Wheat	10		10	14	Nov-Mar	4
Kharif Maize	4				Jun-Sep	
Soybean	2				Jun-Oct	

[d]Theoretical irrigation demand was used instead of the number of irrigations. Methods and data used are provided in supplementary data.

Table A.9 Number of irrigations by crop and irrigation method for Cluster 5, Vijaynagar, Satara

Crop	Average number of irrigations	Standard. Deviation	Minimum number of irrigations	Maximum number of irrigations	Season	Number of farmers surveyed
Sugarcane (18 month)	40		18	48	Jun- Dec	6
Sugarcane (12 month)	25		18	27	Jun-May	5
Soybean	0.5		0	1	Jun - Sep	3
Wheat	9		6	8	Nov-Mar	2
Rabi Sorghum	1.5		1	2	Nov-Feb	3
Ginger drip [d]	55		69	105	Jun-Mar	4
Tomato	5				Jun-Nov	FGD

[d]Theoretical irrigation demand was used instead of the number of irrigations. Methods and data used are provided in supplementary data.

Table A.10 Correlation of crops to water source access. 96 farmers were surveyed on their crops and water sources. Each row lists the number of farmers cultivating a particular category of crop and their water source access. Assured water sources refers to borewells in addition to open wells, or long distance pumping from surface water sources.

Crop type	Area covered (Mha)	Number of farmers	Water source access		Water source crops are correlated to	Proportion of crop irrigated by given water source	
			Only open well	Assured water sources		> 0.7	> 0.5
Kharif and long kharif crops	10.7	96	85	11	Open well	**	***
Gram, Rabi sorghum	4.7	30	21	9	Open well		*
Other Rabi crops	1.5	40	10	30	Assured water source		***
Plantation crops, and sugarcane	1.4	23	3	20	Assured water source	*	***

Table A.11 Water source -crop correlation, seasonality of water sources, and transfers in the 5 clusters

Cluster No. (Name, District)		Water source	Crops	Proportion by months	Water Transfers
1 (Umbarda, Washim))		Open well	Khari, gram, rabi sorghum	Borewell usage is 40% in November increasing to 100% in April,	Borewell to open well
		Open well , Borewell	Other Rabi and plantation crops		
2 (Alasana, Buldhana))		Open well	All crops		No transfers
3 (Garaj, Aurangabad)		Open well	Kharif, gram, rabi sorghum	Borewell usage is 50% in November increasing to 80% in May	Borewell to open well
		Open well , Borewell	Other Rabi and plantation crops		
4 (Old Dawachwad, Nasik)		Open well	Kharif	Canal usage is 35% in December increasing to 100% in February	Canal to open well and storage tanks
		Open well , Canal	Other Rabi and plantation crops		
5 (Vijayn agar, Satara)	Up to 0.5 km from the river	River	Other rabi and sugarcane		No transfers
	Further inland	Open well	Kharif		

Table A.12 Summarized results of cluster 1, Umbarda , Washim

	Irrgtd Soybean[d]	Cotton border	Cotton drip	Pigeon Pea	Gram sprinkl-er	Gram border	Wheat	Orange drip	Orange furrow	Vegetables (chilli)
Cropped area (ha)	46	728	243	768	306	76	515	127.5	43	65
Irrigation applied (mm)	97	162	170	104	197	216	515	1477 (2 furrow + 46 l/t/d/)[@]	977	405
Energy usage (kWh/ ha)	243	378	600	269	708	540	3820	4242	10556	1013
Energy Intensity (kWh/ m^3)	0.26	0.24	0.35	0.26	0.33	0.26	0.74	1.04	1.09	0.26
Profit from surveys (USD./ ha)*	280	240 **(400)	240 **(893)	853	200	200	133	8000	8000	1733
Energy cost as a % of profit	17%	19%	12%	3%	31%	27%	277%	5%	13%	6%

Table A.13 Summarized results of cluster 3, Garaj

	Cotton Border	Cotton Drip	Gram Bor-der	Wheat	Rabi Sorghum	Kharif Maize	Sweet Orange Drip	Tomato Furrow	Ginger Drip	Rabi Onion Border	Rabi Onion drip
Cropped area (ha)	213	142	22	20	13	154	28	14	45	18	4
Irrigation applied (mm)	108	34	184	515	286	54	1477 (2 furrow + 46 l/t/d/)[@]	194	629	648	444
Energy usage (kWh/ ha)	281	112	477	3616	744	140	3536	504	5729	7442	4256
Energy Intensity total (kWh/ m^3)	0.26	0.33	0.26	0.70	0.26	0.26	1.14	0.26	0.91	1.15	0.96
Profit from surveys (USD./ ha)[a]	240(400)	240(893)[b]	200	133	27	773	8000	1400	1507	800	800
Energy cost as a % of profit[c]	7%	1%	23%	262%	270%	2%	4%	3%	37%	100%	51%

[@] *The average litres per day per tree is considered over the irrigation period from November to mid-May. There are 500 trees/ha.*

[a] *USD rate INR 75 (29/10/2021)*

[b] *The profits this year are almost halved due to early harvesting of cotton. The numbers in parentheses are from previous years' cost. % of profit is calculated based on previous years profits. Since the number of irrigations (border method) is normally ~6, which is twice of the year under consideration, we have doubled the cost of energy for the % of profit calculation.*

[c] The per unit cost of energy considered is the average cost of supply incurred by the utility, 9.7 ¢/kWh (Maharashtra State Electricity Distribution Company Ltd., 2020)

[d] Most of the soybean is rainfed. Where it says irrigated, 10% of the soybean is irrigated as per the given data

Table A.14 Summarized results of cluster 2, Alasana, Buldhana

	Cotton border	Cotton drip	Irrgtd Soy-bean	Pigeon Pea	Gram	Wheat	Sugarcane (12 mth)
Cropped area (ha)	389	195	115.5	135	86	93	33
Irrigation applied (mm)	270	504	81	90	120	336	1620
Energy usage open well (kWh/ha)	702	1663	211	234	396	920	4212
Energy Intensity total (kWh/m^3)	0.26	0.33	0.26	0.26	0.33	0.27	0.26
Profit (USD / ha)[a]	211	557	72	482	181	117	1030
Energy cost as a % of profit[c]	32%	29%	28%	5%	21%	76%	40%

[a] USD rate INR 75 (29/10/2021)

[c]The per unit cost of energy considered is the average cost of supply incurred by the utility, 9.7 ¢/kWh (Maharashtra State Electricity Distribution Company Ltd., 2020)

Table A.15 Summarized results of cluster 4, Old Dawachwadi, Nasik

	Soybean	Wheat	Kharif Maize	Rabi Onion	Grape Drip
Cropped area (ha)	71	101	178	38	415
Irrigation applied mm	108	540	180	540	1288 [d]
Energy usage (kWh/ ha)	270	2260	450	2475	5007
Energy Intensity total (kWh/m^3)	0.26	0.41	0.26	0.40	0.41
Profit (USD / ha) [a]	72	117	47	667	14,000
Energy cost as a % of profit[c]	38%	187%	96%	36%	4%

[a] USD rate INR 75 (29/10/2021)

[d] Irrigation applied estimated using theoretical irrigation demand

[c]The per unit cost of energy considered is the average cost of supply incurred by the utility, 9.7 /kWh (Maharashtra State Electricity Distribution Company Ltd., 2020)

Table A.16 Summarized results of cluster 5, Vijaynagar , Satara

	Soy-bean	Wheat	Rb. Sorghum	Sugarcane (18 mth)	Sugarcane (12 mth)	Ginger	Tomato
Cropped area (ha)	363	37	320	282	38	80	60
Irrigation applied (mm)	27	486	81	2160	1350	817[d]	225
Energy usage (kWh/ha)	70	1264	211	5616	3510	2696	585
Energy Intensity total (kWh/m^3)	0.26	0.26	0.26	0.26	0.26	0.33	.26
Profit (USD / ha) [a]	72	117	111	2,730	1,428	1506	1400
Energy cost as a % of profit[c]	9%	104%	18%	20%	24%	19%	4%

[a] USD rate INR 75 (29/10/2021)

[d] Irrigation applied estimated using theoretical irrigation demand

[c]The per unit cost of energy considered is the average cost of supply incurred by the utility, 9.7 ¢/kWh (Maharashtra State Electricity Distribution Company Ltd., 2020)